

A universe exit does not erase a holding

Daily learning note | 3 October 2026 | Q009b

Sidiq review status: Not yet reviewed.

Research question and hypothesis

Can terminal valuation preserve a held security and distinguish unknown proceeds from an explicit zero? The hypothesis is that separating membership, event evidence and cash valuation prevents an eligibility exit from erasing a position or inventing a loss. Complete cash evidence should permit hand-checked arithmetic; incomplete evidence should remain visibly unresolved.

Q009a established historical membership intervals yesterday. Q009b is the next eligible dependency before lagged features. It adds a bounded retrospective cash-only calculation, not a portfolio ledger. All inputs below are synthetic; no real market data, performance estimate or alpha claim is involved.

Beginner explanation and definitions

Imagine owning a ticket that is removed from an approved shopping list. That removal says what you may buy next; it does not say what your existing ticket is worth or whether someone paid you for it. A security leaving a research universe has the same distinction: eligibility, ownership and proceeds are different records. A delisting may require further evidence before the investment can be valued.

Membership: eligibility under a dated universe rule. **Held position:** an existing quantity of shares, independent of eligibility. **Reference value:** shares times the earlier reference price. **Terminal event:** an asserted event resolving the security under this cash-only contract. **Proceeds per share:** complete cash entitlement per share on the reference position's share basis. **Evidence locator:** a retained reference to the supporting record, not proof the record is authentic.

Unknown: insufficient evidence for complete proceeds; represented by None, not zero. **Explicit zero:** a caller-supported assertion that complete proceeds are zero. **Entitlement:** an amount owed, valued at par here; it is not necessarily cash already received. **Resolved:** arithmetic can be performed under supplied assertions; no settlement or audit is implied.

Data, metrics and evaluation protocol

Use one synthetic identity OLD, 10 shares priced at 20 on January 2, 2026, membership exit on January 5 and possible terminal event on January 7. These are calendar-date labels, not an exchange-session sample. Evaluate six hand states, two wrong controls, invalid inputs, membership independence, exact rational accounting and two subprocess replays. Compare states and missing values exactly; numeric comparisons use relative and absolute tolerances of 1e-12. Run full project checks and verify this PDF's text/layout.

Mathematics and worked example

Let $q > 0$ be shares, $P > 0$ the reference price and $C \geq 0$ complete cash proceeds per reference share. Reference value $B = q * P$. Terminal entitlement $V = q * C$. Fractional return $R = V/B - 1 = (C - P)/P$. Return is dimensionless; B and V are currency amounts. Shares and prices must use a consistent basis. There are no intervening dividends, splits or trades. Missing C makes V and R unknown.

For $q=10$ and $P=20$, $B=200$. With $C=5$, $V=50$ and $R=50/200-1=-0.75$, or -75% . The cash payment recovers part of the investment, but it is assumed to be the COMPLETE payment. If 5 is only a known instalment and later payments remain uncertain, use unknown complete proceeds. Changing the number of shares scales the values but leaves the percentage return unchanged.

Evidence case	Terminal value	Fractional return	Result state
No terminal evidence supplied	Unknown	Unknown	missing_event
Event known; complete proceeds missing	Unknown	Unknown	missing_proceeds
Explicit complete zero	0	-1.00	resolved
Complete cash of 5 per share	50	-0.75	resolved
Complete cash of 20 per share	200	0.00	resolved
Complete cash of 25 per share	250	+0.25	resolved

Two deliberately wrong controls

The membership history includes OLD before January 5 and excludes it from that date. Filtering the existing holdings by this new universe produces an empty list. That is wrong: the 10-share position remains part of the audit input. Membership itself gives neither a trade nor a terminal event. The API preserves the supplied position in every result and never takes a membership filter.

A second wrong rule replaces missing cash with zero. It then reports $V=0$ and $R=-100\%$, manufacturing a numeric result from absent evidence. Our unknown cases return no numeric value or return. Conversely, an explicit supported zero must remain valid and yield -100% ; a truthiness check would confuse them.

No event evidence is not proof an event occurred. `missing_event` describes the attempted terminal calculation only. It does not decide the position's current mark. Neither an old carried-forward price nor dropping unresolved holdings is a justified substitute for missing terminal evidence.

Implementation and reproduction

src/quant_research/data/terminal.py adds frozen HeldPosition, TerminalEvidence and TerminalValuation records plus value_terminal. The result retains the position and optional event evidence. IDs and locators reject empty, whitespace and control-character values. Dates reject datetime objects and strings, avoiding silent time loss. The event must match the security and strictly follow priced_on. Same-day ordering is outside this date-only contract and is rejected even when proceeds are unknown.

Shares and reference prices must be positive finite built-in numbers. Proceeds must be explicitly None or a finite nonnegative number. Booleans, strings, negatives and nonfinite values fail. Values are normalized to floats; oversized integers, product/ratio overflow and underflow to zero fail. Frozen fields prevent ordinary assignment, not deliberate low-level modification of Python objects.

scripts/terminal_experiment.py writes experiments/terminal/results.json with inputs, evidence states, values, preserved shares and wrong controls. Run make terminal-experiment, make check and make note NOTE_DATE=2026-10-03. The experiment README contains the full contract and limitations.

Observed results and validation

Check	Observed result
Hand evidence states	All six match; values unknown, unknown, 0, 50, 200, 250
Membership independence	OLD exits eligibility; the 10-share audit position remains
Exact Fraction ledger	108 comparisons agree across quantities, prices, cash and scales
Targeted suite	70 new tests pass, including identity, dates, missingness and range
Reproduction	Two subprocess runs match archived JSON bytes
Full project	540 Python tests; 66 C++ checks and golden output pass
Code quality	Ruff lint/44 formatted files; strict mypy on 20 source files pass

The independent ledger computes exact debit, credit and profit before dividing profit by debit. It covers fractional shares, explicit zero and gains/losses under three currency scales. A fresh C++17 build, dependency consistency and all four original Q003 provenance hashes pass. Initial full checking found one 101-character script line; it was wrapped and the full check then passed. No coverage percentage, mutation-test result or empirical conclusion is claimed.

Primary sources consulted

CRSP payment categories distinguish cash, stock/property, missing information and declared worthlessness. CRSP aggregate flags distinguish missing and partial values. These motivate explicit evidence states; our complete-cash contract does not implement CRSP codes or vendor return conventions. Accessed October 3.

https://www.crsp.org/wp-content/uploads/appendix/FlagType_PT.html

https://www.crsp.org/wp-content/uploads/appendix/FlagType_DE.html

Python dataclasses documents emulated frozen fields; runtime validation remains our responsibility.
<https://docs.python.org/3/library/dataclasses.html#frozen-instances>

Limitations and interview explanation

The caller asserts complete cash terms, identity, entitlement, currency and share basis. A syntactically valid locator cannot authenticate any assertion. Partial payments or stock/property consideration cannot be represented by a cash component alone; keep complete proceeds unknown. No automatic mapping from a delisting code to zero is implemented. Q004a licensed sample evidence remains a prerequisite.

This model is retrospective. An event date does not prove when information became available; there is no announcement time, revision selection, latency, vintage or settlement date. Q005b must connect provenance and availability before empirical use. Do not use the result directly as a past feature. We value complete cash entitlement at par without discounting, credit risk, costs or tax. No portfolio aggregation or position settlement is implemented; unresolved positions must not silently disappear.

Floats are approximate, not exact money. Extreme products may fail even if an alternative algebraic form would avoid that intermediate range problem. Tiny positive recovery can produce a return rounded to -1 during subtraction even when its terminal value is positive. Inspect the value and evidence state; do not infer proven worthlessness from a rounded percentage. No speed or performance claim is made.

In an interview, distinguish eligibility, holdings, event evidence and valuation. Explain why a membership exit cannot create a sale, demonstrate the 200-to-50 example, then contrast unknown with supported zero. Finish with the missing controls: authentic complete evidence, information timing and an explicit settlement ledger. Repository implementation is separate from personal independent mastery.

Next work and necessary input

Q009b and the bounded Q009 parent are complete. Q010 lagged features is Ready for October 4; Q014 remains the separate Sunday afternoon synthesis. C003 stays Wednesday October 7. No input is needed for synthetic continuation. Real-data work needs existing licensed access or a source/budget preference, then a sample audit; do not send credentials. Personal review remains Not yet reviewed.

Daily work remains at 08:00 Europe/London including daylight-saving changes. Stop new research after 31 December 2026 and request the next scope. Publication authorisation does not imply personal review.

Check your understanding

1. What happens to a held security when it leaves an eligible universe?
2. What return follows from 10 shares at 20 and complete terminal cash of 5 per share?
3. Why are unknown proceeds and explicitly supported zero different?
4. Can a dated terminal evidence record safely become a historical trading feature?

Suggested answers

1. Membership alone does not sell, delete or value it. Preserve the position until supported events and ledger rules determine the accounting change.
2. Reference value is 200, terminal entitlement is 50 and the fractional return is -0.75, or -75%.
3. Unknown produces no numeric terminal value or return. Supported complete zero yields 0 and -100%. A partial instalment is not complete proceeds; uncertainty about the remainder must stay unresolved.
4. No. Effective event dates alone do not establish publication time, revisions or what was known at the decision. Provenance and availability selection must be added before empirical feature use.